

Artificially Intelligent

Group 1: Genson, Chien Shyong, Zachary, John-David, Noven

Introduction

With harmful speech, such as racism, cyberbullying, and other targeted abuse, growing in scale and complexity, there is a pressing need for tools that can automatically identify and classify such content.

Following the “AI for Good” theme, we aim to train an AI model to distinguish between various forms of harmful speech, supporting data-driven efforts in digital safety, moderation, and social research.



Cyberbullying on X

X was identified as the second most common platform for online harassment, with 27% of surveyed individuals reporting experiences of cyberbullying on the platform.

The problem has only gotten worse with the downsizing of the moderation team at X.



Problem Statement

This study addresses the challenge by developing an AI classification model that leverages advanced natural language processing (NLP) techniques to detect and categorise cyberbullying within tweets.



Dataset

Type: Cyberbullying Classification

Source: Compiled by IEEE for the SOSNet study on Graph Convolutional Networks

Availability: Publicly available on Kaggle

Size: 46017 manually labelled tweets

Labels: Age, Ethnicity, Gender, Religion, other type of cyberbullying, not cyberbullying

Distribution: Equally distributed across all six classes

△ tweet_text Text of the tweet	△ cyberbullying_type Type of cyberbullying harassment.
46017 unique values	religion17% age17% Other (31702)66%
In other words #katandandre, your food was crapilicious! #mkr	not_cyberbullying
Why is #aussietv so white? #MKR #theblock #ImACelebrityAU #today #sunrise #studio10 #Neighbours #Won...	not_cyberbullying

Data Visualisation

TF-IDF (Term Frequency–Inverse Document Frequency)

- Helps identify the most distinctive words for each class

Purpose:

- Highlights important terms within each cyberbullying category
- Downweights commonly used words that appear across multiple classes

Outcome:

- Extracted top terms per class with their TF-IDF scores

	age	ethnicity	gender	not_cyberbullying	other_cyberbullying	religion
0	school (0.686)	n[REDACTED] (0.590)	r[REDACTED]e (0.547)	mkr (0.684)	rt (0.424)	muslims (0.443)
1	high (0.420)	f[REDACTED]k (0.443)	gay (0.493)	http (0.394)	http (0.401)	muslim (0.376)
2	bullied (0.363)	dumb (0.442)	jokes (0.435)	rt (0.290)	just (0.270)	idiot (0.277)
3	bully (0.228)	n[REDACTED]s (0.207)	joke (0.246)	bullying (0.266)	bully (0.263)	christian (0.241)
4	girl (0.195)	ass (0.193)	rt (0.157)	just (0.164)	https (0.262)	islamic (0.230)
5	girls (0.194)	tayyoung_ (0.183)	people (0.128)	bully (0.151)	like (0.225)	idiots (0.210)
6	like (0.153)	black (0.179)	sexist (0.124)	like (0.147)	don (0.210)	radical (0.195)
7	bullies (0.135)	white (0.130)	just (0.121)	don (0.127)	people (0.204)	like (0.194)
8	just (0.090)	rt (0.104)	like (0.117)	school (0.126)	f[REDACTED]king (0.176)	islam (0.193)
9	people (0.077)	people (0.100)	women (0.115)	amp (0.104)	mkr (0.157)	terrorism (0.182)

Pre-Processing

Negation Handling

- Preserve meaning of phrases like “not happy”
- Convert negations into forms that models can recognise

Tokenisation

- Split text into meaningful units (words, punctuation, emojis)
- Prepares input for downstream NLP tasks

Lemmatisation

- Reduce words to base or root form (e.g., “running” → “run”)
- Retains grammatical and contextual meaning

Normalisation Techniques

- Convert text to lowercase for consistency
- Standardise spellings and punctuation
- Replace slang and contractions (e.g., “u” → “you”, “don’t” → “do not”)

Models Used

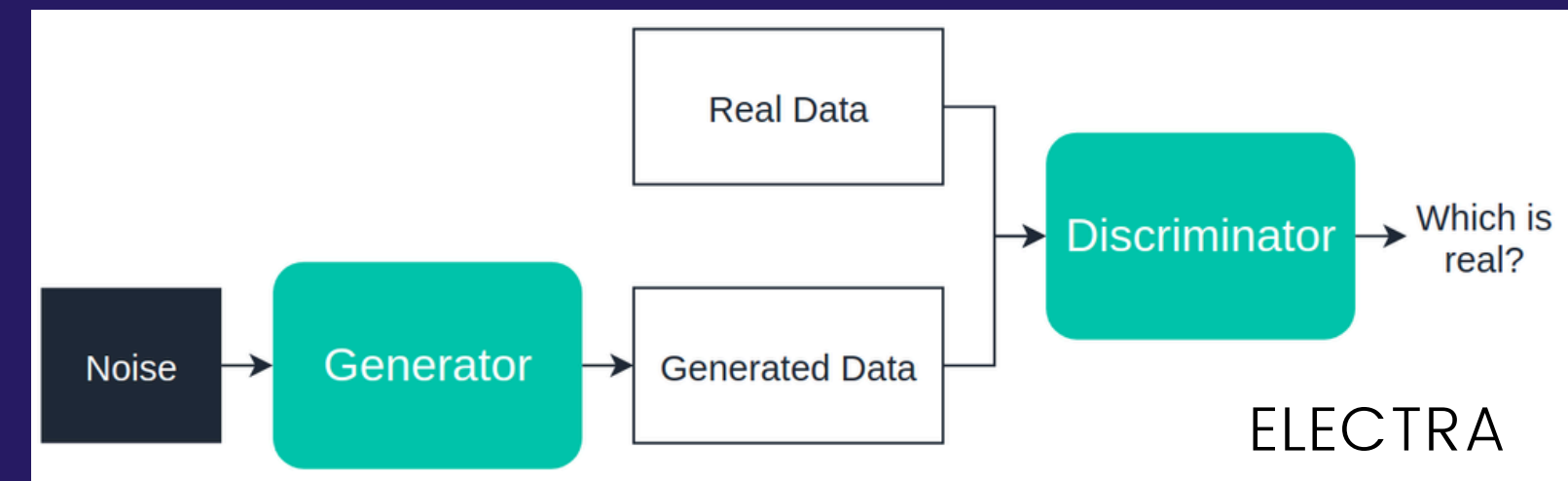
ELECTRA – Detects if tokens are replaced by a generator instead of predicting masked tokens

DeBERTa – Separates word content and position during attention

RoBERTa – An improved BERT variant trained longer with dynamic masking and larger batches

Random Forest – An ensemble of decision trees that aggregates predictions via majority voting

Stacking – Combines predictions from DeBERTa, ELECTRA, and Random Forest using a logistic regression meta-model for improved overall accuracy



Model Training

Common Settings for Transformer Models

- Tokenization with padding/truncation (max length: 128)
- Loss Function: CrossEntropyLoss
- Optimiser: AdamW (learning rate: $2e-5$)
- Training: 4 epochs, batch size: 16

DeBERTa (microsoft/deberta-v3-base)

- Fine-tuned using a custom PyTorch training loop

ELECTRA (google/electra-base-discriminator)

- Fine-tuned using Hugging Face Trainer API

RoBERTa (facebook/roberta-base)

- Also trained using Hugging Face Trainer API

Random Forest Classifier

- Trained on 25 engineered features (structural, lexical, sentiment, POS-based)

Stacking

- Loss Function: LogLoss
- Optimiser: Limited-memory Broyden–Fletcher–Goldfarb–Shanno algorithm

Results

DeBERTa

	precision	recall	f1-score	support
age	0.99	0.98	0.99	1598
ethnicity	0.98	0.97	0.98	1592
gender	0.86	0.92	0.89	1595
not_cyberbullying	0.74	0.61	0.67	1589
other_cyberbullying	0.71	0.79	0.75	1565
religion	0.96	0.96	0.96	1600
accuracy			0.87	9539
macro avg	0.87	0.87	0.87	9539
weighted avg	0.87	0.87	0.87	9539

Electra

	precision	recall	f1-score	support
age	0.98	0.98	0.98	1627
ethnicity	0.99	0.99	0.99	1641
gender	0.89	0.90	0.90	1629
not_cyberbullying	0.70	0.61	0.65	1516
other_cyberbullying	0.68	0.77	0.72	1561
religion	0.96	0.97	0.97	1565
accuracy			0.87	9539
macro avg	0.87	0.87	0.87	9539
weighted avg	0.87	0.87	0.87	9539

Results

RoBERTa

	precision	recall	f1-score	support
not_cyberbullying	0.76	0.57	0.65	1624
other_cyberbullying	0.70	0.78	0.74	1612
age	0.99	0.97	0.98	1603
religion	0.96	0.96	0.96	1566
gender	0.80	0.94	0.87	1531
ethnicity	0.98	0.98	0.98	1603
accuracy			0.87	9539
macro avg	0.87	0.87	0.86	9539
weighted avg	0.87	0.87	0.86	9539

Random Forests

	precision	recall	f1-score	support
age	0.67	0.69	0.68	1598
ethnicity	0.83	0.66	0.74	1592
gender	0.66	0.60	0.63	1590
not_cyberbullying	0.45	0.50	0.47	1587
other_cyberbullying	0.46	0.52	0.49	1565
religion	0.65	0.66	0.65	1600
accuracy			0.61	9532
macro avg	0.62	0.61	0.61	9532
weighted avg	0.62	0.61	0.61	9532

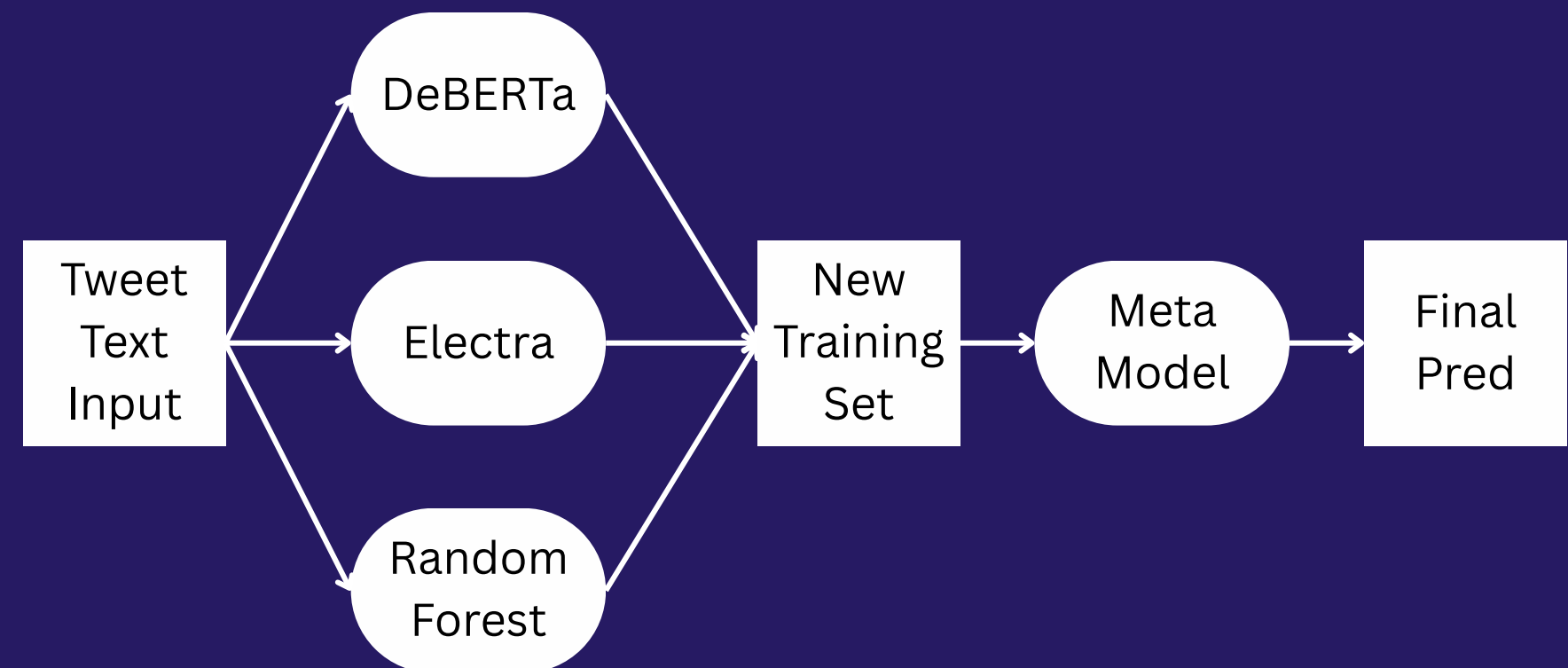
Ensemble: Stacking

Ensemble Learning:

- Combine low-performing models into one better-performing model

Stacking:

- Combines outputs from DeBERTa, ELECTRA, and Random Forest
- Final classifier optimised using log loss, solved via L-BFGS optimiser
- Improving overall accuracy and robustness.



Results

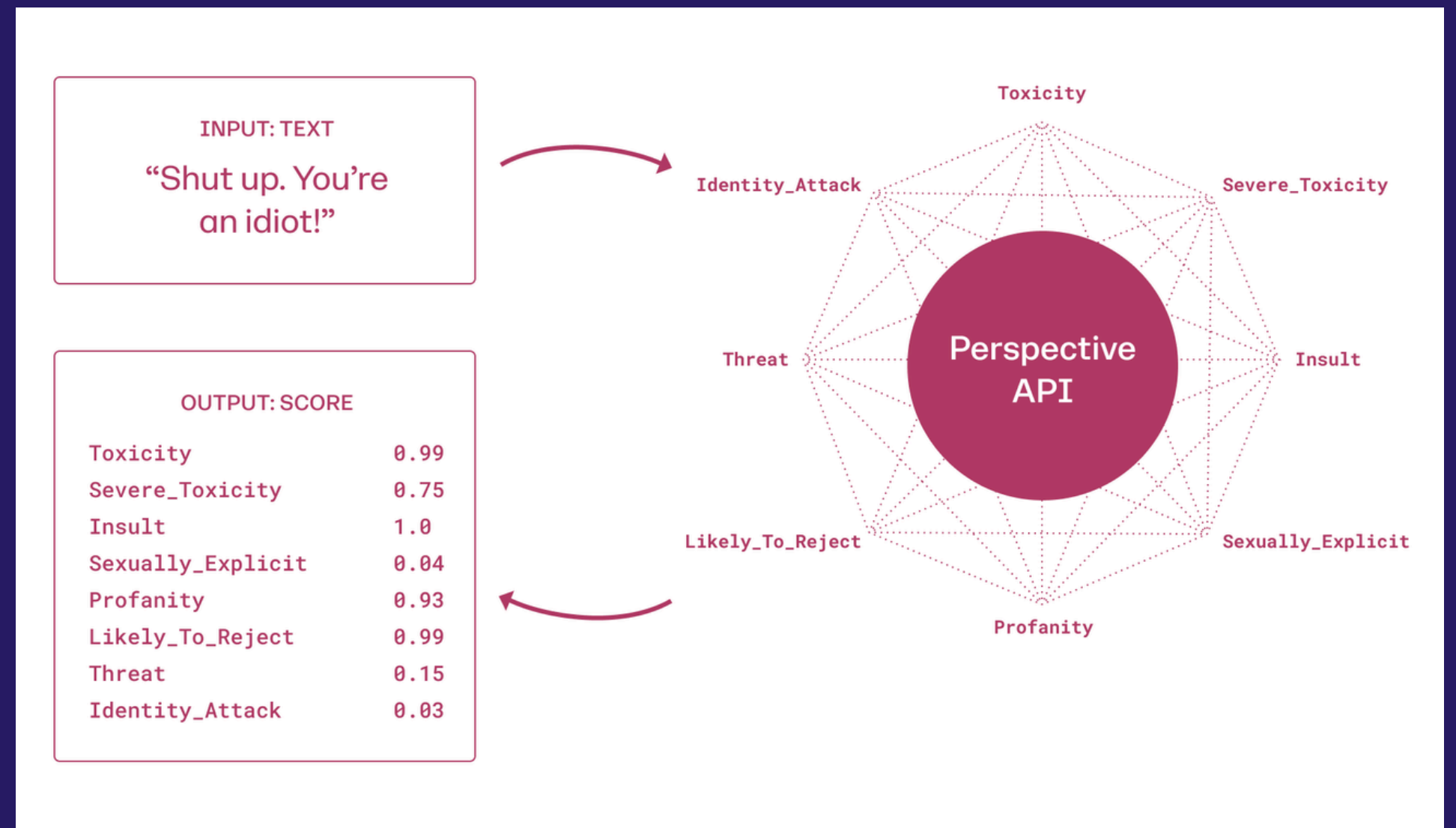
Stacking (DeBERTa, Electra, Random Forest)

	precision	recall	f1-score	support
age	0.99	0.99	0.99	1598
ethnicity	0.99	0.97	0.98	1592
gender	0.93	0.95	0.94	1595
not_cyberbullying	0.66	0.81	0.73	1589
other_cyberbullying	0.75	0.58	0.66	1565
religion	0.98	0.98	0.98	1600
accuracy			0.88	9539
macro avg	0.89	0.88	0.88	9539
weighted avg	0.89	0.88	0.88	9539

Baseline Comparison

Google Jigsaw's Perspective API

- The API provides a score between 0 (not toxic) and 1 (highly toxic).
- Websites can use threshold-based filtering (e.g., flagging comments with scores above 0.7).



Baseline Comparison

Google Jigsaw's Perspective API

We record the toxicity score returned by the API (0.00 to 1.00), then, we classify the tweets as toxic or not, experimenting with different thresholds:

`thresholds = [i / 100 for i in range(10, 91)]`

The best threshold is 0.18 with an F1 score of 0.7617.

Challenges

Ambiguity in Tweets

- Subtle cues like sarcasm and irony complicate detection.
- Misclassification is common in "other_cyberbullying" vs "not_cyberbullying".

Model Complexity and Resource Demands

- Ensemble models (e.g., RoBERTa, DeBERTa) improve accuracy but are computationally intensive.

Dataset Inconsistencies

- Manual labels can vary due to subjective judgment
- Inconsistent annotations affect model reliability and generalisability.

```
data[data['cleaned_words']=='']
```

	tweet_text	cyberbullying_type	cleaned_words
307	:D	not_cyberbullying	
2675	That is all	not_cyberbullying	
4011	♪♥♪	not_cyberbullying	
4016	#M...	not_cyberbullying	
4582	Why?	not_cyberbullying	
6451	No. Just no.	not_cyberbullying	
6545	Just NOW?!?!? 🤔🤔🤔	not_cyberbullying	
8695	🤔🤔🤔🤔🤔	gender	
9981	🤔 no it isn't	gender	
14458	🤔🤔🤔🤔 ...	gender	
14691	♪♪	gender	
24110	?????????	other_cyberbullying	
26069	🤔🤔🤔🤔🤔🤔🤔	other_cyberbullying	
26604	WHAT IS THIS.	other_cyberbullying	
28045	My.	other_cyberbullying	
28437	NO!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!!...	other_cyberbullying	
28468	:~)	other_cyberbullying	
28855	🤔	other_cyberbullying	
31286	.	other_cyberbullying	

Conclusion

The growing volume of online content makes it difficult to efficiently understand public sentiment on societal issues.

This project aims to develop an AI-powered sentiment analysis model to provide insights, enabling better informed decision-making for positive social impact.



Demo

 Cyberbullying Classifier

Enter text to classify:

Choose Model:

DeBERTa

Predict

Thank You

Question & Answer